Definite Integration (2)

Q1. Find $\lim_{a\to\infty} \int_0^1 e^{x^2} \cos ax\, dx$

Q2. For $a>0$, $b>0$, verify that $\int_0^\infty \frac{\ln x\, dx}{ax^2+bx+a}$ reduces to zero by a substitution $x=\frac{1}{t}$. Using this or otherwise, evaluate $\int_0^\infty \frac{\ln x\, dx}{x^2+2x+4}$.

Q3. Comment upon the nature of roots of the quadratic equation $x^2+2x = k+\int_0^1 |t+k|\,dt$ depending on the value of $k\in R$.

Q4. Let f be a continuous function on $[a,b]$. If $F(x)=\left(\int_a^x f(t)\,dt - \int_x^b f(t)\,dt\right)(2x-(a+b))$ then prove that there exist some $c\in(a,b)$ such that $\int_a^c f(t)\,dt - \int_c^b f(t)\,dt = f(c)(a+b-2c)$.

Q5. If f is a continuous real function such that $f(x-1)+f(x+1) \geq 1+f(x)$ for all x, what is the minimum value of $\int_1^{2005} f(x)\,dx$?

Q6. Let $F:(1,\infty)\to R$ be the function defined by $F(x)=\int_x^{x^2} \frac{dx}{\ln t}$. Show that F is one-one and find the range of F.

Q7. Let f be a continuous function on $[a,b]$ such that for every $x\in[0,1]$ we have $\int_0^1 f(t)\,dt \geq \frac{1-x^2}{2}$. Show that $\int_0^1 f^2(t)\,dt \geq \frac{1}{3}$.

Q8. Evaluate the limit $\lim_{x\to 0+} \int_x^{2x} \frac{\sin^m t}{t^n}\,dt$ $(m,n\in N)$

Q9. Let $\langle \epsilon_n \rangle$ be a sequence of positive real numbers, such that $\lim_{n\to\infty} \epsilon_n = 0$. Find $\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \ln\left(\frac{k}{n}+\epsilon_n\right)$.

Q10. Prove that $\int_0^1 x^{n-1}(1-x)^{m-1}\,dx = \frac{1\cdot 2\cdot 3\cdots(m-1)}{n(n+1)\cdots(n+m-1)}$ when m and n are positive and m is an integer.

Q11. Let $f(x)$, $x \geq 0$ be a non-negative continuous function, and let $F(x) = \int_0^x f(t)\,dt$, $x \geq 0$. If for some $c > 0$, $f(x) \leq cF(x)$ for all $x \geq 0$, then show that $f(x) = 0$ for all $x \geq 0$.

Q12. If $a, b, c \in R^+$ then show that $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{n}{(k+an)(k+bn)}$ is equal to (i) $\frac{1}{a-b}\ln\frac{a(b+1)}{b(a+1)}$ if $a \neq b$ (ii) $\frac{1}{a(1+a)}$ if $a = b$.

Q13. Evaluate $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{n-k}{n^2} \cos\frac{4k}{n}$.

Q14. Let $a + b = 4$ where $a < 2$ and $g(x)$ is a differentiable function of x. If $\frac{dg}{dx} > 0$ for all x, prove that $\int_0^a g(x)\,dx + \int_0^b g(x)\,dx$ increases as $(b-a)$ increases.

Q15. For a positive constant t, let α, β be the roots of the quadratic equation $x^2 + t^2x - 2t = 0$. If the minimum value of $\int_{-1}^{2}\left[\left(x+\frac{1}{\alpha^2}\right)\left(x+\frac{1}{\beta^2}\right)+\frac{1}{\alpha\beta}\right]dx$ is $\sqrt{\frac{a}{b}} + c$, where $a, b, c \in N$, then find the least value of $(a+b+c)$.

Q16. If $f(x) = a|\cos x| + b|\sin x|$ $(a, b \in R)$ has a local minimum at $x = -\frac{\pi}{3}$ and satisfies $\int_{-\pi/2}^{\pi/2} (f(x))^2\,dx = 2$. Find the values of a and b and hence find $\frac{b^2}{a^2}$.

Q17. If the value of definite integral $\int_e^{e^{2010}} \frac{1}{x}\left[1 + \frac{1-\ln x}{\ln x \ln\left(\frac{x}{\ln x}\right)}\right]dx = A - \ln(4 - \ln 4)$ then evaluate $(A - 2)$.

Q18. Let $F(x) = \int_{-1}^{x} \sqrt{4+t^2}\,dt$ and $G(x) = \int_{x}^{1} \sqrt{4+t^2}\,dt$; then compute the value of $(FG)'(0)$.

Q19. Let $u = \int_0^{\pi/4} \left(\frac{\cos x}{\sin x + \cos x}\right)^2 dx$ and $v = \int_0^{\pi/4} \left(\frac{\sin x + \cos x}{\cos x}\right) dx$, find the value of $\frac{v}{u}$.

Q20. Evaluate $\lim_{n\to\infty} n^2 \int_{-1/n}^{1/n} (2006\sin x + 2007\cos x)\,|x|\,dx$

Q21. If a_1, a_2, a_3 are the three values of a which satisfy the equation $\int_0^{\pi/2} (\sin x + a\cos x)^3 dx - \frac{4a}{\pi-2}\int_0^{\pi/2} x\cos x\,dx = 2$, then find the value of $1000(a_1^2 + a_2^2 + a_3^2)$.

Q22. If α, β be the distinct positive roots of the equation $\tan x = 2x$, then evaluate $\int_0^1 (\sin\alpha x \cdot \sin\beta x)\,dx$, independent of α & β.

Q23. Let $f(x) = \sqrt{x}$ and $g(x) = e^x - 1$ then $\int_0^x fog(x)\,dx = A\,fog(x) + B\tan^{-1}(fog(x))$, then evaluate $(A-B)$.

Q24. If p is the number of points of extrema of the function $\int_4^{x^2} \frac{t(t-1)(t-9)^2}{et-1}\,dt$ and $q = \int_{10}^{20} \left\{\frac{x}{15}\right\} dx$; where $\{.\}$ denotes fractional part function and $r = \int_0^{2\pi} \ln\left(\sin x + \sqrt{1+\sin^2 x}\right) dx$, then evaluate $p+q+r$.

Q25. A function f is defined for all real x satisfying the equation $f(x) = 5 + \int_0^x \frac{1+f(t)}{2+t^2}\,dt$. Find a quadratic polynomial $p(x) = a + bx + cx^2$ such that $p(0) = f(0)$, $p'(0) = f'(0)$ & $p''(0) = f''(0)$. Hence find $p(2)$.

P: (There exists a G.P with first term and its common ratio (say α & β, respectively) satisfying the equation $ax^2 + bx + a = 0$ for $a, b \in R$. on adding 1 to twice the sum of n terms of G.P. we get the quantity equal to the number of ways of selecting objects out of n different object. Further, on interchanging the first term and common ratio of G.P. the sum of new obtained infinite G.P becomes 7 and we have $3\int_{\beta-2}^{\alpha-2} (x+1)^n\,dx = 364$.

Q26. $\int_0^{\pi/4} \sin^n(2x)\,dx$ equals

(a) $\frac{15}{16}$ (b) $\frac{21\pi}{15}$ (c) $\frac{4}{15}$ (d) NOT

Q27. $\int_0^{\pi/2} \sin^{2\beta} x \cos^{2\alpha} x\,dx$ equals

(a) $\frac{7\pi}{2^{11}}$ (b) $\frac{7}{2^{10}}$ (c)

Q28. The value of $\int_0^{3\pi} x\,|\sin^6 x \cos^n x|\,dx$ equals

(a) $\frac{2}{5}$ (b) $\frac{5}{2}$ (c) $\frac{5\pi}{4}$ (d) NOT.

P: Let $f(x)$ be a polynomial of the form x^n for some $n \in N$ and it one-one and onto. Let $\bar{f}^1(x)$ be the inverse of $f(x)$. Further define a function $g(x) = \min\{f(x), \bar{f}^1(x)\}$ such that

$\lim_{x\to 1}\left[\frac{g(x)-x}{h(x)}+\frac{|x-1|}{x^2+2x-3}\right]$ is non-zero finite for least possible value of x, and $h(x)$ is a second degree polynomial even function.

Q 29. The value of $\int_1^2 h(x)\,dx$ is equal to

(a) $\frac{24}{9}$ (b) $\frac{32}{9}$ (c) $\frac{32}{3}$ (d) NOT.

Q 30. The value of $\int_0^3 g(x)\,dx$ equals

(a) $\frac{3}{4}((3)^{4/3}-1)$ (b) $\frac{1}{4}((3)^{4/3}-1)$

(c) $\frac{1}{4}((3)^{7/3}-1)$ (d) NOT.

Q 31. The value of $\int_0^L \frac{g(x)-x}{h(x)}\,dx$ (where L is the given limit) is

(a) $\frac{3}{1024}$ (b) $\frac{3}{512}$ (c) $\frac{5}{256}$ (d) NOT.

Q 32. Show that $\int_0^{p+q\pi}|\cos x|\,dx = 2q+\sin p$, where $q\in N$ and $-\frac{\pi}{2}<p<\frac{\pi}{2}$.

Q 33. $\int_1^{16}\tan^{-1}\left(\sqrt{\sqrt{x}-1}\right)dx$

Q 34. If $\int_0^x\sqrt{(\cos x+\cos 2x+\cos 3x)^2+(\sin x+\sin 2x+\sin 3x)^2}\,dx$ has the value equal to $\left(\frac{\pi}{k}+\sqrt{\omega}\right)$, where k & ω are positive integers, then find the value of $(k^2+\omega^2)$.

Q 35. $\int_0^{\pi/2} e^x\left\{\cos(\sin x)\cos^2\frac{x}{2}+\sin(\sin x)\sin^2\frac{x}{2}\right\}dx$

Q 36. Let f be a function continuous on R satisfying the equation

$\int_0^x f(t)\,dt = \int_0^t t^2 f(t)\,dt + \frac{x^{16}}{8}+\frac{x^{18}}{8}+c$

Find the function $f(x)$ and the constant c.

Q37. If f is a continuous function with $\int_0^x f(t)\,dt \to \infty$ as $|x| \to \infty$, then show that every line $y = mx$ intersects the curve $y^2 + \int_0^x f(t)\,dt = 2$.

Q38. Compute the following integrals

a) $\int_0^\infty f(x^n + x^{-n}) \ln x \frac{dx}{x} = 0$

b) $\int_0^\infty f(x^n + x^{-n}) \ln x \frac{dx}{1+x^2} = 0$

Q39. If f, g, h be continuous function on $[0, a]$ such that $f(a-x) = f(x)$, $g(a-x) = g(x)$ and $3h(x) - 4h(a-x) = 5$, then prove that $\int_0^a f(x) g(x) dx = -\frac{1}{5} \int_0^a f(x) g(x) h(x) dx$

Q40. Evaluate $\int_0^\infty [ne^{-x}]\,dx$, where n is a natural number greater than 1 and $[\cdot]$ is GIF.

Q41. Evaluate $\int_0^\pi \frac{(ax+b)\sec x \tan x}{4 + \tan^2 x}\,dx$

Q42. Evaluate $\int_0^{2\pi} \frac{\sin 2\theta}{a - b\cos\theta}\,d\theta \quad (a > b > 0)$

Q43. $f(x) = -\int_0^x \log \cos y\,dy$; prove that

$f(x) = 2f\left(\frac{\pi}{4} + \frac{x}{2}\right) - 2f\left(\frac{x}{4} - \frac{x}{2}\right) - x\log 2$.

Hence or otherwise, show that $f\left(\frac{\pi}{2}\right) = \frac{\pi}{2}\log 2$

Q44. The value of $\lim_{n\to\infty} \left(\frac{n!}{n^n}\right)\left(\frac{2n^4+1}{5n^5+1}\right)$.

Q45. If the value of $\int_0^\pi \left(\frac{x}{1+\sin x}\right)^2 dx = I$, then find the value of the integral $\int_0^\pi \left[\frac{2x^2 \cos^2(x/2)}{(1+\sin x)^2}\right] dx$.

Q46. $\int_0^\infty \frac{(x-1)^4}{(x^8+x^{10})}\,dx$

Q47. $\int_0^\pi \frac{\log(1-t^2\sin^2 x)}{\sin x}\,dx = -2(\sin^{-1}t)^2,\ 0\le t\le 1$

Q48. $\int_0^1 \frac{dx}{(5+2x-2x^2)(1+e^{(2-4x)})}\,dx$

Q49. If $f(x)$ is a given function and $f'(x)$ is continuous for all x, find y for which $\frac{d^2y}{dx^2} = \int_{-x}^{x} f(t)\,dt$.

Q50. For what values of 'a' the following statement is true

$\int_0^\pi \frac{dx}{a^2-2a\cos x+1} = \frac{\pi}{a^2-1}\operatorname{sgn}(a-1)$.

Q51. $\int_0^{\pi/6} \frac{\sqrt{3\cos 2x-1}}{\cos x}\,dx$

Q52. $\int_0^\pi \frac{x^2\sin 2x\,\sin\left(\frac{\pi}{2}\cos x\right)}{(2x-\pi)}\,dx = \frac{8}{\pi}$.

Q53. A sequence of non-negative terms is given by $a_{n+1} = \int_{\frac{2}{3}(a_{n-1})}^{a_n} 3\,dx$ where $a_0=0$, $a_1=1$. Then the value $(a_1+a_2+a_3+\cdots+a_{100})$ equals

(A) 2^{100} (B) $2^{100}-100$ (C) $2^{100}-101$ (D) $2^{101}-102$.

Q54. If f is continuous function and $F(x) = \int_0^x \left((2t+3)\cdot\int_t^2 f(u)\,du\right)dt$ then $F''(2)$ is equal to

(A) $-7f(2)$ (B) $7f'(2)$ (C) $3f'(2)$ (D) $7f(2)$.

Q55. Let x_1 and x_2 $(x_1<x_2)$ be two values of x satisfying $||x|^2-4|x|+5|+||x^3-4x|+3| = 4$ and $f(x)$ is a continuous periodic function with period $\frac{1}{2}$ such that $\int_0^{3/2} f(x)\,dx = 6$. Find the value of $\int_{x_1}^{x_2} f(x)\,dx$.

Q56. If value of $\int_0^{\ln 2} \frac{(x-\ln 2)e^{x+\ln 2}}{e^{\ln(1+e^x)^2}}\,dx$ can be expressed in the form $\ln\left(\frac{a}{b}\right)$ where $a, b \in N$ and are relatively prime, then find the value of $a+b$.

Q57. If the value of $\int_{1/3}^{1/2} \{x\left[\frac{2}{x}\right]\}\,dx = \frac{m}{n}$ where m & n are coprime then find the absolute value of $(m-n)$.
(where $[\cdot]$ & $\{\cdot\}$ GIF)

Q58. Let $p(x)$ be fifth degree polynomial such that $p(x)+1$ is divisible by $(x-1)^3$ and $p(x)-1$ is divisible by $(x+1)^3$. Then find the value of $\int_{-10}^{10} p(x)\,dx$.

Q59. Let $Fn = \int \frac{x^n}{e^x}\,dx$, then find the value of ~~(Q56 F16 7 I14)~~ $\int_2^5 \left(e^{\int \frac{F(1)}{F(2)}\,dx}\right)^2$ where $F(1)$ and $F(2)$ contains no constant term. Also $\int \frac{F(1)}{F(2)}\,dx$ contains no constant term.

Q60. Let $f(x) = ax^3+bx^2+cx+d$ $(a,b,c,d \in R)$ satisfy the following three conditions
i) $f(1)=1$ ii) $f(-1)=-1$ iii) $\int_{-1}^{1}(bx^2+cx+d)\,dx = 1$.

Q61. Let the function $f(x)$ and $g(x)$ are differentiable and strictly increasing $\forall$ $x \in R$. Also $f(0)=2$, $f(1)=3$ and $f'(x)$ is continuous. If $f^2(x) = 1+g^2(x)$ and $J = \int_0^1 \frac{f(x)g'(x)-f'(x)g(x)}{f^2(x)g(x)}\,dx$, then find the value of $6J+1$ if equal to $\ln\frac{a}{b}$, $a, b \in N$. Find the minimum value of $a+b$.

Q62. If $\lim_{t\to\infty}\left(\int_1^t \frac{dx}{1+\sqrt[3]{x^2}} - at + b\right)$ exists and equals to non-zero finite number L, where a & b are positive real numbers and $t>1$, then find $(ab - 4L - 3\pi)$.

Q63. Let f be a function defined by
$f\left(\frac{x}{x-1}\right) = \frac{1}{x}$ $\forall$ $x \in R - \{0,1\}$.
If $f(\sec 2\theta)$ is equal to $g(\theta)$ where $\theta \in (0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi)$ then the value of $\int_0^{\pi} \frac{d\theta}{1+2g(\theta)}$ is $\frac{\pi}{m}$ $(m \in R)$.
Find the value of m^2.

Q64. Consider $f(x) = 3\int_0^1 x^2 f(t)\,dt + 2\int_0^1 x\, f(|1-t|)\,dt + 3$.

Q65. If $f(x) = \lim_{t\to 0} \int_0^x \left[\frac{tx^2 + x\left((t+1)^{2t+2} - 1\right)}{e^x t}\right] dx$
then find the number of solutions of the equation $2f(-|x|) - 1 = 0$